However, a larger radius for the core density areas mean that these areas will contain more points and this makes the algorithm slower.

In the main section of the algorithm, in fact, for each point processed we need to update the reachability distance of all undiscovered points in its ε-neighborhood: the more points it contains, the slower the algorithm will be.

As we briefly mentioned, OPTICS main cycle grows current cluster[22] C by adding the closest point to the cluster's frontier, which is the point with the smallest reachability distance from any point already in C; the cluster is formed in a way similar to DBSCAN's because all points reachable from the cluster's seed (the first point processed for a new cluster) are added to it. However, forming these clusters is not the direct objective of OPTICS.

When a new point is "discovered" and processed, its reachability distance from the cluster is set in stone. This cluster-to-point distance is not to be confused with the reachability distance between two points. For a point q and a cluster C, we define the reachability distance of q from C as the minimum of the reachability distance from p to q, for all points p in C:

$$\texttt{reachability-distance}_{\varepsilon,M}(q,C) = \\ min\left\{\texttt{reachability-distance}_{\varepsilon,M}(q,p)\,\forall p \mid p\,\epsilon\,C\right\}$$

If when we process a point p we update the reachability distances of all points in its ε-neighborhood, and we keep a